

FDA Patient Summaries and Quiz

https://gemini.google.com/app/8f2e6f7d50475c66

User prompt: hmm remember each visit is a seperate tab

Response: **TAB 1: Patient Profile & Medical History**

- **Subject ID:** MRN 7167
- **Demographics:** 40-year-old African woman.
- **Obstetric History:** Gravida 4, Para 2 (2 previous live births via C-section, 1 miscarriage). Previous C-sections were indicated due to a macrosomic baby and hypertension.
- **Chronic Conditions:** Pre-gestational Diabetes Mellitus and Chronic Hypertension.
- **Concomitant Medications:** Actively managed on Methyldopa (500mg tds), Metformin/Ranophage (1g bd), Insulin Actrapid (8IU tds), Insulin Protaphane (12IU bd), and Aspirin (150mg od).
- **Family History:** Positive for both Hypertension and Diabetes.

 TAB 2: Visit 1 (Screening) - 12/May/2026

- **Gestational Age:** 30 weeks, 3 days
- **Vitals:** BP 114/78 mmHg | HR 90 bpm | Temp 36.6°C | Weight 82.8 kg

Category	Test	Result	Interpretation
Urinalysis	Blood	Negative	Normal
	Ketones	Negative	Normal
	Leukocytes	1+	High
	Nitrites	Negative	Normal
	Protein	Negative	Normal
	Glucose	Negative	Normal
Haematology	Haemoglobin	11.41 g/dL	Normal
	Haematocrit	33.6%	Low
	WBC Count	6.40 x10 ⁹ /L	Normal
	RBC Count	3.62 x10 ¹² /L	Low
	Platelets	251.5 x10 ⁹ /L	Normal
	ANC	4.54 x10 ⁹ /L	Normal
Biochemistry	ALC	1.39 x10 ⁹ /L	Low
	Creatinine	33.7 µmol/L	Low
	ALT	9.2 U/L	Normal
	AST	11.4 U/L	Normal
	ALP	90.9 U/L	Normal
	Total Bilirubin	6.32 µmol/L	Normal
	LDH	139.6 U/L	Low

 TAB 3: Visit 2 (Follow-up) - 21/May/2026

- **Gestational Age:** 31 weeks, 5 days
- **Clinical Status:** Clinically stable. No new adverse events.
- **Vitals:** BP 128/85 mmHg (Recheck: 127/86) | HR 78 bpm | Temp 36.7°C | Weight 83.0 kg

Category	Test	Result	Interpretation
Urinalysis	Blood	Negative	Normal
	Ketones	Negative	Normal
	Leukocytes	Negative	Normal
	Nitrites	Negative	Normal
	Protein	Negative	Normal
	Glucose	Negative	Normal
Haematology	Haemoglobin	11.86 g/dL	Normal
	Haematocrit	34.5%	Low
	WBC Count	6.58 x10 ⁹ /L	Normal
	RBC Count	3.76 x10 ¹² /L	Low
	Platelets	240.8 x10 ⁹ /L	Normal
	ANC	4.92 x10 ⁹ /L	Normal
	ALC	1.19 x10 ⁹ /L	Low
Biochemistry	Creatinine	44.24 µmol/L	Low
	ALT	5.38 U/L	Normal
	AST	11.86 U/L	Normal
	ALP	92.79 U/L	Normal
	Total Bilirubin	5.29 µmol/L	Normal
	LDH	154.57 U/L	Low

 TAB 4: Visit 3 (Follow-up) - 27/May/2026

- **Gestational Age:** 32 weeks, 4 days
- **Clinical Status:** Clinically stable. No medication changes.
- **Vitals:** BP 132/89 mmHg (Recheck: 129/88) | HR 82 bpm | Temp 36.6°C | Weight 83.0 kg

Category	Test	Result	Interpretation
Urinalysis	Blood	Negative	Normal
	Ketones	Trace	High
	Leukocytes	Negative	Normal
	Nitrites	Negative	Normal
	Protein	Trace	High
	Glucose	Negative	Normal
Haematology	Haemoglobin	11.97 g/dL	Normal
	Haematocrit	35.2%	Low
	WBC Count	6.57 x10 ⁹ /L	Normal
	RBC Count	3.81 x10 ¹² /L	Normal
	Platelets	223.0 x10 ⁹ /L	Normal
	ANC	5.03 x10 ⁹ /L	Normal
	ALC	1.14 x10 ⁹ /L	Low
Biochemistry	Creatinine	36.74 µmol/L	Low

Category	Test	Result	Interpretation
	ALT	7.23 U/L	Normal
	AST	11.55 U/L	Normal
	ALP	91.32 U/L	Normal
	Total Bilirubin	5.60 µmol/L	Normal
	LDH	162.74 U/L	Low

 TAB 5: Visit 4 (Follow-up) - 11/Jun/2026

- **Gestational Age:** ~34 weeks, 5 days
- **Clinical Status:** Clinically stable leading into scheduled delivery window.
- **Vitals:** BP 129/78 mmHg (Recheck: 125/75) | HR 90 bpm | Temp 36.4°C | Weight 82.0 kg

Category	Test	Result	Interpretation
Urinalysis	Blood	Negative	Normal
	Ketones	Negative	Normal
	Leukocytes	Negative	Normal
	Nitrites	Negative	Normal
	Protein	Negative	Normal
	Glucose	Negative	Normal
Haematology	Haemoglobin	11.60 g/dL	Normal
	Haematocrit	34.0%	Low
	WBC Count	6.50 x10 ⁹ /L	Normal
	RBC Count	3.67 x10 ¹² /L	Low
	Platelets	249.0 x10 ⁹ /L	Normal
	ANC	4.50 x10 ⁹ /L	Normal
	ALC	1.57 x10 ⁹ /L	Normal
Biochemistry	Creatinine	38.01 µmol/L	Low
	ALT	4.95 U/L	Low
	AST	10.0 U/L	Normal
	ALP	109.7 U/L	Normal
	Total Bilirubin	6.38 µmol/L	Normal
	LDH	152.97 U/L	Low

 TAB 6: Visit 5 (Delivery) - 29/Jun/2026

- **Gestational Age at Delivery:** 37 weeks, 3 days
- **Delivery Outcome:** Elective Cesarean Section (Indication: 2 Previous CS).
- **Infant Status:** Live normal birth. Apgar scores: 8 (1 min), 9 (5 mins).
- **Maternal Status:** Normal postpartum recovery; estimated blood loss 200mL. No blood or urine tests were mandated for this visit.

 TAB 7: Visit 6 (Postnatal) - 28/Jul/2026

- **Gestational Age:** Postpartum (~4 weeks)
- **Clinical Status:** Unremarkable postnatal recovery.

- **Vitals:** BP 138/88 mmHg (Recheck: 136/89) | HR 88 bpm | Temp 36.2°C | Weight 77.0 kg

Category	Test	Result	Interpretation
Urinalysis	-	Not Collected	-
Haematology	Haemoglobin	13.8 g/dL	Normal
	Haematocrit	40.8%	Normal
	WBC Count	4.38 x10 ⁹ /L	Normal
	RBC Count	1.3 x10 ¹² /L	Low
	Platelets	271.2 x10 ⁹ /L	Normal
	ANC	2.77 x10 ⁹ /L	Normal
	ALC	1.30 x10 ⁹ /L	Low
Biochemistry	Creatinine	53.2 µmol/L	Normal
	ALT	16.9 U/L	Normal
	AST	14.1 U/L	Normal
	ALP	183.2 U/L	High
	Total Bilirubin	6.93 µmol/L	Normal
	LDH	191.9 U/L	Normal

 TAB 8: Adjudicator Summary (ISSHP Criteria Focus)

Clinical Course: A 40-year-old multimorbid patient (G4P2) with established chronic hypertension and pre-gestational diabetes successfully completed her pregnancy, delivering via planned C-section at 37 weeks. Her chronic conditions were medically managed throughout the trial using Methyldopa, Metformin, dual-therapy Insulin, and Aspirin prophylaxis.

Preeclampsia (PE) Diagnostic Evaluation: Under ISSHP guidelines, adjudicating *superimposed preeclampsia* in a patient with chronic hypertension requires evidence of new-onset maternal organ dysfunction or a sudden, severe exacerbation of baseline hypertension.

1. **Hemodynamics:** While her BP trended upward marginally in the third trimester and postnatally (peaking at 138/89 mmHg), it remained well-controlled on Methyldopa and never breached the severe threshold (≥160/110 mmHg) or the standard diagnostic threshold (≥140/90 mmHg) during clinic vitals checks.
2. **Organ Dysfunction:** The patient maintained excellent baseline renal function (creatinine consistently low/normal) and showed no signs of hepatic injury (ALT/AST normal throughout). Platelets remained robustly normal, ruling out HELLP syndrome.
3. **Proteinuria:** She exhibited only a solitary "Trace" protein result at Visit 3, which resolved entirely by Visit 4.

Adjudicator Verdict: The patient exhibits well-controlled chronic hypertension and diabetes without clinical or laboratory evidence of superimposed preeclampsia or uteroplacental failure during the monitored windows.

User prompt: when i say each visit is a tab i mean for every visit there should when one click on v1, they should get dropdowns on the summary the one you had done well , then sepearate dropdowns for patient history and labs (also i am not seeing the gestational age and sonographis data (if present))

Response: Here is the interactive, dropdown-based formatting for each visit. I have integrated HTML <details> and <summary> tags to create clickable tabs, and I've ensured **Gestational Age** and **Sonographic Data** (ultrasounds) are prominently featured under the Patient History/Clinical Status dropdowns where applicable.

(Note: In most Markdown viewers, you can click on the triangles below to expand or collapse each section).

Category	Test	Result	Interpretation
Urinalysis	Leukocytes	1+	High
	Protein/Glucose/Ketones	Negative	Normal
Haematology	Haemoglobin	11.41 g/dL	Normal
	Haematocrit	33.6%	Low
	RBC Count	3.62 x10 ¹² /L	Low

Category	Test	Result	Interpretation
	WBC / Platelets	6.40 / 251.5	Normal
Biochemistry	Creatinine	33.7 µmol/L	Low
	ALT / AST	9.2 / 11.4 U/L	Normal
	LDH	139.6 U/L	Low

Category	Test	Result	Interpretation
Urinalysis	All Parameters	Negative	Normal
Haematology	Haemoglobin	11.86 g/dL	Normal
	Haematocrit	34.5%	Low
	RBC Count	3.76 x10 ¹² /L	Low
	WBC / Platelets	6.58 / 240.8	Normal
Biochemistry	Creatinine	44.24 µmol/L	Low
	ALT / AST	5.38 / 11.86 U/L	Normal
	LDH	154.57 U/L	Low

Category	Test	Result	Interpretation
Urinalysis	Protein	Trace	High
	Ketones	Trace	High
	Leukocytes / Glucose	Negative	Normal
Haematology	Haemoglobin	11.97 g/dL	Normal
	Haematocrit	35.2%	Low
	RBC Count	3.81 x10 ¹² /L	Normal
	Platelets	223.0 x10 ⁹ /L	Normal
Biochemistry	Creatinine	36.74 µmol/L	Low
	ALT / AST	7.23 / 11.55 U/L	Normal
	LDH	162.74 U/L	Low

Category	Test	Result	Interpretation
Urinalysis	All Parameters	Negative	Normal
Haematology	Haemoglobin	11.60 g/dL	Normal
	Haematocrit	34.0%	Low
	RBC Count	3.67 x10 ¹² /L	Low
	Platelets	249.0 x10 ⁹ /L	Normal
Biochemistry	Creatinine	38.01 µmol/L	Low
	ALT / AST	4.95 / 10.0 U/L	Normal (ALT Low)
	LDH	152.97 U/L	Low

Category	Test	Result	Interpretation
Urinalysis	-	Not Collected	-

Category	Test	Result	Interpretation
Haematology	Haemoglobin	13.8 g/dL	Normal
	Haematocrit	40.8%	Normal
	RBC Count	1.3 x10 ¹² /L	Low
	Platelets	271.2 x10 ⁹ /L	Normal
Biochemistry	Creatinine	53.2 µmol/L	Normal
	ALT / AST	16.9 / 14.1 U/L	Normal
	ALP	183.2 U/L	High
	Total Bilirubin	6.93 µmol/L	Normal
	LDH	191.9 U/L	Normal

Clinical Course: A 40-year-old multimorbid patient (G4P2) with established chronic hypertension and pre-gestational diabetes successfully completed her pregnancy, delivering via planned C-section at 37 weeks. Her chronic conditions were medically managed throughout the trial using Methyldopa, Metformin, dual-therapy Insulin, and Aspirin prophylaxis.

Preeclampsia (PE) Diagnostic Evaluation: Under ISSHP guidelines, adjudicating *superimposed preeclampsia* in a patient with chronic hypertension requires evidence of new-onset maternal organ dysfunction or a sudden, severe exacerbation of baseline hypertension.

- 1. Hemodynamics:** While her BP trended upward marginally in the third trimester and postnatally (peaking at 138/89 mmHg), it remained well-controlled on Methyldopa and never breached the severe threshold (≥160/110 mmHg) or the standard diagnostic threshold (≥140/90 mmHg) during clinic vitals checks.
- 2. Organ Dysfunction:** The patient maintained excellent baseline renal function (creatinine consistently low/normal) and showed no signs of hepatic injury (ALT/AST normal throughout). Platelets remained robustly normal, ruling out HELLP syndrome.
- 3. Proteinuria:** She exhibited only a solitary "Trace" protein result at Visit 3, which resolved entirely by Visit 4.

Adjudicator Verdict: The patient exhibits well-controlled chronic hypertension and diabetes without clinical or laboratory evidence of superimposed preeclampsia or uteroplacental failure during the monitored windows.

User prompt: we are now reverting from where we started why are you changing when i simply make an addition Visit 1: Baseline / Screening (12/May/2026) A 40-year-old African woman, Gravida 4, Para 2 (two previous live births via C-section, one miscarriage), presents for her initial clinical trial screening. Her medical history is significant for chronic hypertension and pre-gestational diabetes, actively managed on Methyldopa, Metformin, dual-therapy Insulin, and Aspirin. She reports a family history of both hypertension and diabetes. Gestational Age: 30 weeks, 3 days Vitals: BP 114/78 mmHg | HR 90 bpm | Temp 36.6°C | Weight 82.8 kg CategoryTestResultInterpretationUrinalysisBloodNegativeNormalKetonesNegativeNormalLeukocytes1+HighNitritesNegativeNormalProteinNegativeN g/dLNormalHaematocrit33.6%LowWBC Count6.40 x10⁹/LNormalRBC Count3.62 x10¹²/LLowPlatelets251.5 x10⁹/LNormalANC4.54 x10⁹/LNormalALC1.39 x10⁹/LLowBiochemistryCreatinine33.7 µmol/LLowALT9.2 U/LNormalAST11.4 U/LNormalALP90.9 U/LNormalTotal Bilirubin6.32 µmol/LNormalLDH139.6 U/LLow we had approved this and the additions i stated after

Response: Ah, my apologies! I completely see what you mean. I over-engineered the formatting inside the dropdowns and ended up changing the clean text layout we had already perfected.

Let's strip it right back. Here is the exact structure and text flow you approved, integrated into the dropdowns, with the gestational age and sonographic data seamlessly added in.

A 40-year-old African woman, Gravida 4, Para 2 (two previous live births via C-section, one miscarriage), presents for her initial clinical trial screening. Her medical history is significant for chronic hypertension and pre-gestational diabetes, actively managed on Methyldopa, Metformin, dual-therapy Insulin, and Aspirin. She reports a family history of both hypertension and diabetes.

Gestational Age: 30 weeks, 3 days **Vitals:** BP 114/78 mmHg | HR 90 bpm | Temp 36.6°C | Weight 82.8 kg **Sonographic Data:** (Scan Date: 27/Jan/2026) EGA at scan 15 weeks, 3 days. Fetal weight 120g. Cephalic presentation. Fetal heart rate 159 bpm. Normal amniotic fluid and placenta.

Category	Test	Result	Interpretation
Urinalysis	Blood	Negative	Normal
	Ketones	Negative	Normal
	Leukocytes	1+	High
	Nitrites	Negative	Normal
	Protein	Negative	Normal

Category	Test	Result	Interpretation
	Glucose	Negative	Normal
Haematology	Haemoglobin	11.41 g/dL	Normal
	Haematocrit	33.6%	Low
	WBC Count	6.40 x10 ⁹ /L	Normal
	RBC Count	3.62 x10 ¹² /L	Low
	Platelets	251.5 x10 ⁹ /L	Normal
	ANC	4.54 x10 ⁹ /L	Normal
	ALC	1.39 x10 ⁹ /L	Low
Biochemistry	Creatinine	33.7 µmol/L	Low
	ALT	9.2 U/L	Normal
	AST	11.4 U/L	Normal
	ALP	90.9 U/L	Normal
	Total Bilirubin	6.32 µmol/L	Normal
	LDH	139.6 U/L	Low

The patient returns for her first routine follow-up assessment. She remains clinically stable with no new adverse events reported, and her chronic conditions remain well-controlled on her baseline pharmacological regimen.

Gestational Age: 31 weeks, 5 days **Vitals:** BP 128/85 mmHg (Recheck: 127/86) | HR 78 bpm | Temp 36.7°C | Weight 83.0 kg **Sonographic Data:** (Scan Date: 19/May/2026) EGA at scan 31 weeks. Fetal heart rate 140 bpm. Normal amniotic fluid and placenta. Fetal growth within normal limits.

Category	Test	Result	Interpretation
Urinalysis	All Parameters	Negative	Normal
Haematology	Haemoglobin	11.86 g/dL	Normal
	Haematocrit	34.5%	Low
	WBC Count	6.58 x10 ⁹ /L	Normal
	RBC Count	3.76 x10 ¹² /L	Low
	Platelets	240.8 x10 ⁹ /L	Normal
	ANC	4.92 x10 ⁹ /L	Normal
	ALC	1.19 x10 ⁹ /L	Low
Biochemistry	Creatinine	44.24 µmol/L	Low
	ALT	5.38 U/L	Normal
	AST	11.86 U/L	Normal
	ALP	92.79 U/L	Normal
	Total Bilirubin	5.29 µmol/L	Normal
	LDH	154.57 U/L	Low

At her 32-week follow-up, the patient continues her routine antenatal monitoring. Her hemodynamic profile remains stable without any acute symptomatic exacerbations of her chronic hypertension or diabetes.

Gestational Age: 32 weeks, 4 days **Vitals:** BP 132/89 mmHg (Recheck: 129/88) | HR 82 bpm | Temp 36.6°C | Weight 83.0 kg **Sonographic Data:** No new scan performed since last visit.

Category	Test	Result	Interpretation
Urinalysis	Protein	Trace	High

Category	Test	Result	Interpretation
	Ketones	Trace	High
	Leukocytes	Negative	Normal
	Nitrites	Negative	Normal
	Glucose	Negative	Normal
Haematology	Haemoglobin	11.97 g/dL	Normal
	Haematocrit	35.2%	Low
	WBC Count	6.57 x10 ⁹ /L	Normal
	RBC Count	3.81 x10 ¹² /L	Normal
	Platelets	223.0 x10 ⁹ /L	Normal
	ANC	5.03 x10 ⁹ /L	Normal
	ALC	1.14 x10 ⁹ /L	Low
Biochemistry	Creatinine	36.74 µmol/L	Low
	ALT	7.23 U/L	Normal
	AST	11.55 U/L	Normal
	ALP	91.32 U/L	Normal
	Total Bilirubin	5.60 µmol/L	Normal
	LDH	162.74 U/L	Low

The patient presents for her late third-trimester follow-up leading into her scheduled delivery window. Her health status remains compliant and stable. The previous trace proteinuria has completely resolved.

Gestational Age: ~34 weeks, 5 days **Vitals:** BP 129/78 mmHg (Recheck: 125/75) | HR 90 bpm | Temp 36.4°C | Weight 82.0 kg **Sonographic Data:** No new scan performed since last visit.

Category	Test	Result	Interpretation
Urinalysis	All Parameters	Negative	Normal
Haematology	Haemoglobin	11.60 g/dL	Normal
	Haematocrit	34.0%	Low
	WBC Count	6.50 x10 ⁹ /L	Normal
	RBC Count	3.67 x10 ¹² /L	Low
	Platelets	249.0 x10 ⁹ /L	Normal
	ANC	4.50 x10 ⁹ /L	Normal
	ALC	1.57 x10 ⁹ /L	Normal
Biochemistry	Creatinine	38.01 µmol/L	Low
	ALT	4.95 U/L	Low
	AST	10.0 U/L	Normal
	ALP	109.7 U/L	Normal
	Total Bilirubin	6.38 µmol/L	Normal
	LDH	152.97 U/L	Low

The patient was admitted for an elective Cesarean section indicated by her history of two previous operative deliveries. She delivered a live, healthy infant, and her immediate postpartum recovery was entirely normal.

Gestational Age at Delivery: 37 weeks, 3 days **Delivery Outcome:** Elective Cesarean Section. **Infant Status:** Live normal birth. Apgar scores: 8 (1 min), 9 (5 mins). **Maternal Status:** Normal postpartum recovery. Estimated blood loss: 200mL. No post-delivery infections or readmission.

(No blood or urine tests were mandated per protocol for this delivery visit.)

Approximately four weeks postpartum, the patient returns for her final study follow-up. Her postnatal clinical recovery has been unremarkable. Blood pressure remains slightly elevated but consistent with her chronic baseline.

Gestational Age: Postpartum (~4 weeks) **Vitals:** BP 138/88 mmHg (Recheck: 136/89) | HR 88 bpm | Temp 36.2°C | Weight 77.0 kg **Sonographic Data:** Not applicable.

Category	Test	Result	Interpretation
Urinalysis	-	Not Collected	-
Haematology	Haemoglobin	13.8 g/dL	Normal
	Haematocrit	40.8%	Normal
	WBC Count	4.38 x10 ⁹ /L	Normal
	RBC Count	1.3 x10 ¹² /L	Low
	Platelets	271.2 x10 ⁹ /L	Normal
	ANC	2.77 x10 ⁹ /L	Normal
	ALC	1.30 x10 ⁹ /L	Low
Biochemistry	Creatinine	53.2 µmol/L	Normal
	ALT	16.9 U/L	Normal
	AST	14.1 U/L	Normal
	ALP	183.2 U/L	High
	Total Bilirubin	6.93 µmol/L	Normal
	LDH	191.9 U/L	Normal

Clinical Course: A 40-year-old multimorbid patient (G4P2) with established chronic hypertension and pre-gestational diabetes successfully completed her pregnancy, delivering via planned C-section at 37 weeks. Her chronic conditions were medically managed throughout the trial using Methyldopa, Metformin, dual-therapy Insulin, and Aspirin prophylaxis.

Preeclampsia (PE) Diagnostic Evaluation: Under ISSHP guidelines, adjudicating *superimposed preeclampsia* in a patient with chronic hypertension requires evidence of new-onset maternal organ dysfunction or a sudden, severe exacerbation of baseline hypertension.

1. **Hemodynamics:** While her BP trended upward marginally in the third trimester and postnatally (peaking at 138/89 mmHg), it remained well-controlled on Methyldopa and never breached the severe threshold (≥160/110 mmHg) or the standard diagnostic threshold (≥140/90 mmHg) during clinic vitals checks.
2. **Organ Dysfunction:** The patient maintained excellent baseline renal function (creatinine consistently low/normal) and showed no signs of hepatic injury (ALT/AST normal throughout). Platelets remained robustly normal, ruling out HELLP syndrome.
3. **Proteinuria:** She exhibited only a solitary "Trace" protein result at Visit 3, which resolved entirely by Visit 4.

Adjudicator Verdict: The patient exhibits well-controlled chronic hypertension and diabetes without clinical or laboratory evidence of superimposed preeclampsia or uteroplacental failure during the monitored windows.

User prompt: can we try v5 and 6 as they include delivery and post delivery visit

Response: Here are Visits 5 and 6 isolated and formatted with the exact dropdown structure. I've enriched the **Patient History** for the delivery visit to ensure all the critical delivery and newborn metrics (hospital stay, blood loss, Apgar scores) are cleanly captured.

The patient was admitted for an elective Cesarean section indicated by her history of two previous operative deliveries. She delivered a live, healthy infant without any intraoperative or postpartum complications, and recovered normally before being discharged.

Gestational Age at Delivery: 37 weeks, 3 days **Delivery Outcome:** Elective Cesarean Section **Primary Indication:** 2 Previous CS **Maternal Postpartum Status:** Normal recovery.

- **Estimated Blood Loss:** 200 mL
- **Post-delivery Infection:** None
- **Hospital Stay:** 2 days (No readmission) **Infant Status:** Live normal birth.
- **Apgar Scores:** 8 (at 1 min), 9 (at 5 mins)

(As per the trial protocol, routine blood and urine tests were not mandated or collected for the delivery module.)

Approximately four weeks postpartum, the patient returns for her final study follow-up. Her postnatal clinical recovery has been unremarkable. Blood pressure remains slightly elevated but is completely consistent with her chronic baseline hypertension.

Gestational Age: Postpartum (~4 weeks) **Vitals:** BP 138/88 mmHg (Recheck: 136/89) | HR 88 bpm | Temp 36.2°C | Weight 77.0 kg **Clinical Status:** Unchanged, clinically stable. **Sonographic Data:** Not clinically indicated postpartum.

Category	Test	Result	Interpretation
Urinalysis	-	Not Collected	-
Haematology	Haemoglobin	13.8 g/dL	Normal
	Haematocrit	40.8%	Normal
	WBC Count	4.38 x10 ⁹ /L	Normal
	RBC Count	1.3 x10 ¹² /L	Low
	Platelets	271.2 x10 ⁹ /L	Normal
	ANC	2.77 x10 ⁹ /L	Normal
	ALC	1.30 x10 ⁹ /L	Low
Biochemistry	Creatinine	53.2 µmol/L	Normal
	ALT	16.9 U/L	Normal
	AST	14.1 U/L	Normal
	ALP	183.2 U/L	High
	Total Bilirubin	6.93 µmol/L	Normal
	LDH	191.9 U/L	Normal

Let me know if this hits the mark or if you want to tweak the delivery metrics breakdown!

User prompt: bp should be there

Response: You're completely right to expect blood pressure at the delivery visit! However, looking closely at the extracted data for **Visit 5 (29/Jun/2026)**, the site coordinators only completed the "Delivery Outcome", "Newborn Assessment", and "Conmeds + AE Assessment" pages in the CTMS. They **did not** capture or export a "Vital Signs" page for the delivery date in this specific file.

Blood pressure resumes in the dataset at **Visit 6 (28/Jul/2026)**.

Here are Visits 5 and 6 using the exact dropdown layout you approved:

The patient was admitted for an elective Cesarean section indicated by her history of two previous operative deliveries. She delivered a live, healthy infant without any intraoperative or postpartum complications, and recovered normally before being discharged.

Gestational Age at Delivery: 37 weeks, 3 days **Vitals:** (Not captured/exported in the CTMS for the delivery date). **Delivery Outcome:** Elective Cesarean Section (Indication: 2 Previous CS). **Maternal Postpartum Status:** Normal recovery. Estimated Blood Loss: 200 mL. Hospital Stay: 2 days (No post-delivery infection or readmission). **Infant Status:** Live normal birth. Apgar scores: 8 (at 1 min), 9 (at 5 mins).

(As per the trial protocol, routine blood and urine tests were not mandated or collected for the delivery module.)

Approximately four weeks postpartum, the patient returns for her final study follow-up. Her postnatal clinical recovery has been unremarkable. Blood pressure remains slightly elevated but is completely consistent with her chronic baseline hypertension.

Gestational Age: Postpartum (~4 weeks) **Vitals:** BP 138/88 mmHg (Recheck: 136/89) | HR 88 bpm | Temp 36.2°C | Weight 77.0 kg **Clinical Status:** Unchanged, clinically stable. **Sonographic Data:** Not clinically indicated postpartum.

Category	Test	Result	Interpretation
Urinalysis	-	Not Collected	-
Haematology	Haemoglobin	13.8 g/dL	Normal
	Haematocrit	40.8%	Normal
	WBC Count	4.38 x10 ⁹ /L	Normal
	RBC Count	1.3 x10 ¹² /L	Low

Category	Test	Result	Interpretation
	Platelets	271.2 x10 ⁹ /L	Normal
	ANC	2.77 x10 ⁹ /L	Normal
	ALC	1.30 x10 ⁹ /L	Low
Biochemistry	Creatinine	53.2 µmol/L	Normal
	ALT	16.9 U/L	Normal
	AST	14.1 U/L	Normal
	ALP	183.2 U/L	High
	Total Bilirubin	6.93 µmol/L	Normal
	LDH	191.9 U/L	Normal

User prompt: okay thats cool now make a complete of all visits and i also need fetal assesment notes for v5

Response: Here is the complete, compiled record for MRN 7167 across all six visits, featuring the newly expanded fetal and newborn assessment notes for Visit 5.

A 40-year-old African woman, Gravida 4, Para 2 (two previous live births via C-section, one miscarriage), presents for her initial clinical trial screening. Her medical history is significant for chronic hypertension and pre-gestational diabetes, actively managed on Methyldopa, Metformin, dual-therapy Insulin, and Aspirin. She reports a family history of both hypertension and diabetes.

Gestational Age: 30 weeks, 3 days **Vitals:** BP 114/78 mmHg | HR 90 bpm | Temp 36.6°C | Weight 82.8 kg **Sonographic Data:** (Scan Date: 27/Jan/2026) EGA at scan 15 weeks, 3 days. Fetal weight 120g. Cephalic presentation. Fetal heart rate 159 bpm. Normal amniotic fluid and placenta.

Category	Test	Result	Interpretation
Urinalysis	Blood	Negative	Normal
	Ketones	Negative	Normal
	Leukocytes	1+	High
	Nitrites	Negative	Normal
	Protein	Negative	Normal
	Glucose	Negative	Normal
Haematology	Haemoglobin	11.41 g/dL	Normal
	Haematocrit	33.6%	Low
	WBC Count	6.40 x10 ⁹ /L	Normal
	RBC Count	3.62 x10 ¹² /L	Low
	Platelets	251.5 x10 ⁹ /L	Normal
	ANC	4.54 x10 ⁹ /L	Normal
	ALC	1.39 x10 ⁹ /L	Low
Biochemistry	Creatinine	33.7 µmol/L	Low
	ALT	9.2 U/L	Normal
	AST	11.4 U/L	Normal
	ALP	90.9 U/L	Normal
	Total Bilirubin	6.32 µmol/L	Normal
	LDH	139.6 U/L	Low

The patient returns for her first routine follow-up assessment. She remains clinically stable with no new adverse events reported, and her chronic conditions remain well-controlled on her baseline pharmacological regimen.

Gestational Age: 31 weeks, 5 days **Vitals:** BP 128/85 mmHg (Recheck: 127/86) | HR 78 bpm | Temp 36.7°C | Weight 83.0 kg **Sonographic Data:** (Scan Date: 19/May/2026) EGA at scan 31 weeks. Fetal heart rate 140 bpm. Normal amniotic fluid and placenta. Fetal growth within normal limits.

Category	Test	Result	Interpretation
Urinalysis	All Parameters	Negative	Normal
Haematology	Haemoglobin	11.86 g/dL	Normal
	Haematocrit	34.5%	Low
	WBC Count	6.58 x10 ⁹ /L	Normal
	RBC Count	3.76 x10 ¹² /L	Low
	Platelets	240.8 x10 ⁹ /L	Normal
	ANC	4.92 x10 ⁹ /L	Normal
	ALC	1.19 x10 ⁹ /L	Low
Biochemistry	Creatinine	44.24 µmol/L	Low
	ALT	5.38 U/L	Normal
	AST	11.86 U/L	Normal
	ALP	92.79 U/L	Normal
	Total Bilirubin	5.29 µmol/L	Normal
	LDH	154.57 U/L	Low

At her 32-week follow-up, the patient continues her routine antenatal monitoring. Her hemodynamic profile remains stable without any acute symptomatic exacerbations of her chronic hypertension or diabetes.

Gestational Age: 32 weeks, 4 days **Vitals:** BP 132/89 mmHg (Recheck: 129/88) | HR 82 bpm | Temp 36.6°C | Weight 83.0 kg **Sonographic Data:** No new scan performed since last visit.

Category	Test	Result	Interpretation
Urinalysis	Protein	Trace	High
	Ketones	Trace	High
	Leukocytes	Negative	Normal
	Nitrites	Negative	Normal
	Glucose	Negative	Normal
Haematology	Haemoglobin	11.97 g/dL	Normal
	Haematocrit	35.2%	Low
	WBC Count	6.57 x10 ⁹ /L	Normal
	RBC Count	3.81 x10 ¹² /L	Normal
	Platelets	223.0 x10 ⁹ /L	Normal
	ANC	5.03 x10 ⁹ /L	Normal
	ALC	1.14 x10 ⁹ /L	Low
Biochemistry	Creatinine	36.74 µmol/L	Low
	ALT	7.23 U/L	Normal
	AST	11.55 U/L	Normal
	ALP	91.32 U/L	Normal
	Total Bilirubin	5.60 µmol/L	Normal

Category	Test	Result	Interpretation
	LDH	162.74 U/L	Low

The patient presents for her late third-trimester follow-up leading into her scheduled delivery window. Her health status remains compliant and stable. The previous trace proteinuria has completely resolved.

Gestational Age: ~34 weeks, 5 days **Vitals:** BP 129/78 mmHg (Recheck: 125/75) | HR 90 bpm | Temp 36.4°C | Weight 82.0 kg **Sonographic Data:** No new scan performed since last visit.

Category	Test	Result	Interpretation
Urinalysis	All Parameters	Negative	Normal
Haematology	Haemoglobin	11.60 g/dL	Normal
	Haematocrit	34.0%	Low
	WBC Count	6.50 x10 ⁹ /L	Normal
	RBC Count	3.67 x10 ¹² /L	Low
	Platelets	249.0 x10 ⁹ /L	Normal
	ANC	4.50 x10 ⁹ /L	Normal
	ALC	1.57 x10 ⁹ /L	Normal
	Basophils	0.00 x10 ⁹ /L	Normal
Biochemistry	Creatinine	38.01 µmol/L	Low
	ALT	4.95 U/L	Low
	AST	10.0 U/L	Normal
	ALP	109.7 U/L	Normal
	Total Bilirubin	6.38 µmol/L	Normal
	LDH	152.97 U/L	Low

The patient was admitted for an elective Cesarean section indicated by her history of two previous operative deliveries. She delivered a live, healthy infant without any intraoperative or postpartum complications, and recovered normally before being discharged.

Gestational Age at Delivery: 37 weeks, 3 days **Vitals:** (Not captured in CTMS for the delivery date). **Delivery Outcome:** Elective Cesarean Section (Indication: 2 Previous CS). **Maternal Postpartum Status:** Normal recovery. Estimated Blood Loss: 200 mL. Hospital Stay: 2 days (No post-delivery infection or readmission). **Fetal/Newborn Assessment Notes:**

- **Status:** Live, normal birth (Male).
- **Anthropometry:** Birth Weight 2550g | Length 45cm | Head Circumference 36cm.
- **Newborn Vitals:** Heart rate 145 bpm | Respiration 41 bpm.
- **Apgar Scores:** 8 (at 1 min), 9 (at 5 mins).
- **Clinical Exams:** Physical and neurological examinations were completely normal (eyes, heart, hips, testes, posture, muscle tone, reflexes). No congenital anomalies, birth complications, or neonatal ICU admission.
- **Feeding:** Breastfeeding established.

(As per the trial protocol, routine blood and urine tests were not mandated or collected for the delivery module.)

Approximately four weeks postpartum, the patient returns for her final study follow-up. Her postnatal clinical recovery has been unremarkable. Blood pressure remains slightly elevated but is completely consistent with her chronic baseline hypertension.

Gestational Age: Postpartum (~4 weeks) **Vitals:** BP 138/88 mmHg (Recheck: 136/89) | HR 88 bpm | Temp 36.2°C | Weight 77.0 kg **Maternal Status:** Unchanged, clinically stable. **Infant Status:** Normal newborn disorder status assessment. No signs of respiratory distress, necrotizing enterocolitis, or intraventricular hemorrhage. No hospitalizations since discharge.

Category	Test	Result	Interpretation
Urinalysis	-	Not Collected	-
Haematology	Haemoglobin	13.8 g/dL	Normal
	Haematocrit	40.8%	Normal

Category	Test	Result	Interpretation
	WBC Count	4.38 x10 ⁹ /L	Normal
	RBC Count	1.3 x10 ¹² /L	Low
	Platelets	271.2 x10 ⁹ /L	Normal
	ANC	2.77 x10 ⁹ /L	Normal
	ALC	1.30 x10 ⁹ /L	Low
Biochemistry	Creatinine	53.2 µmol/L	Normal
	ALT	16.9 U/L	Normal
	AST	14.1 U/L	Normal
	ALP	183.2 U/L	High
	Total Bilirubin	6.93 µmol/L	Normal
	LDH	191.9 U/L	Normal

Clinical Course: A 40-year-old multimorbid patient (G4P2) with established chronic hypertension and pre-gestational diabetes successfully completed her pregnancy, delivering a healthy 2550g male infant via planned C-section at 37 weeks. Her chronic conditions were medically managed throughout the trial using Methyldopa, Metformin, dual-therapy Insulin, and Aspirin prophylaxis.

Preeclampsia (PE) Diagnostic Evaluation: Under ISSHP guidelines, adjudicating *superimposed preeclampsia* in a patient with chronic hypertension requires evidence of new-onset maternal organ dysfunction or a sudden, severe exacerbation of baseline hypertension.

1. **Hemodynamics:** While her BP trended upward marginally in the third trimester and postnatally (peaking at 138/89 mmHg), it remained well-controlled on Methyldopa and never breached the severe threshold (≥160/110 mmHg) or the standard diagnostic threshold (≥140/90 mmHg) during clinic vitals checks.

2. **Organ Dysfunction:** The patient maintained excellent baseline renal function (creatinine consistently low/normal) and showed no signs of hepatic injury (ALT/AST normal throughout). Platelets remained robustly normal, ruling out HELLP syndrome.

3. **Proteinuria:** She exhibited only a solitary "Trace" protein result at Visit 3, which resolved entirely by Visit 4.

Adjudicator Verdict: The patient exhibits well-controlled chronic hypertension and diabetes without clinical or laboratory evidence of superimposed preeclampsia or uteroplacental failure during the monitored windows.

User prompt: *i need visit headings on all 6 like you did for 1*

Response: Got it! You want the visits as clear **Markdown headings** (since your system likely handles the tabs), and *only* the internal sections (Summary, Patient History, Labs) as the clickable dropdowns under each heading.

Here is the complete, correctly formatted record for all 6 visits:

Visit 1: Baseline / Screening (12/May/2026)

A 40-year-old African woman, Gravida 4, Para 2 (two previous live births via C-section, one miscarriage), presents for her initial clinical trial screening. Her medical history is significant for chronic hypertension and pre-gestational diabetes, actively managed on Methyldopa, Metformin, dual-therapy Insulin, and Aspirin. She reports a family history of both hypertension and diabetes.

- **Gestational Age:** 30 weeks, 3 days
- **Vitals:** BP 114/78 mmHg | HR 90 bpm | Temp 36.6°C | Weight 82.8 kg
- **Sonographic Data:** (Scan Date: 27/Jan/2026) EGA at scan 15 weeks, 3 days. Fetal weight 120g. Cephalic presentation. Fetal heart rate 159 bpm. Normal amniotic fluid and placenta.

Category	Test	Result	Interpretation
Urinalysis	Blood	Negative	Normal
	Ketones	Negative	Normal
	Leukocytes	1+	High
	Nitrites	Negative	Normal
	Protein	Negative	Normal

Category	Test	Result	Interpretation
	Glucose	Negative	Normal
Haematology	Haemoglobin	11.41 g/dL	Normal
	Haematocrit	33.6%	Low
	WBC Count	6.40 x10 ⁹ /L	Normal
	RBC Count	3.62 x10 ¹² /L	Low
	Platelets	251.5 x10 ⁹ /L	Normal
	ANC	4.54 x10 ⁹ /L	Normal
	ALC	1.39 x10 ⁹ /L	Low
Biochemistry	Creatinine	33.7 µmol/L	Low
	ALT	9.2 U/L	Normal
	AST	11.4 U/L	Normal
	ALP	90.9 U/L	Normal
	Total Bilirubin	6.32 µmol/L	Normal
	LDH	139.6 U/L	Low

Visit 2: Follow-up (21/May/2026)

The patient returns for her first routine follow-up assessment. She remains clinically stable with no new adverse events reported, and her chronic conditions remain well-controlled on her baseline pharmacological regimen.

- **Gestational Age:** 31 weeks, 5 days
- **Vitals:** BP 128/85 mmHg (Recheck: 127/86) | HR 78 bpm | Temp 36.7°C | Weight 83.0 kg
- **Sonographic Data:** (Scan Date: 19/May/2026) EGA at scan 31 weeks. Fetal heart rate 140 bpm. Normal amniotic fluid and placenta. Fetal growth within normal limits.

Category	Test	Result	Interpretation
Urinalysis	All Parameters	Negative	Normal
Haematology	Haemoglobin	11.86 g/dL	Normal
	Haematocrit	34.5%	Low
	WBC Count	6.58 x10 ⁹ /L	Normal
	RBC Count	3.76 x10 ¹² /L	Low
	Platelets	240.8 x10 ⁹ /L	Normal
	ANC	4.92 x10 ⁹ /L	Normal
	ALC	1.19 x10 ⁹ /L	Low
Biochemistry	Creatinine	44.24 µmol/L	Low
	ALT	5.38 U/L	Normal
	AST	11.86 U/L	Normal
	ALP	92.79 U/L	Normal
	Total Bilirubin	5.29 µmol/L	Normal
	LDH	154.57 U/L	Low

Visit 3: Follow-up (27/May/2026)

At her 32-week follow-up, the patient continues her routine antenatal monitoring. Her hemodynamic profile remains stable without any acute symptomatic exacerbations of her chronic hypertension or diabetes.

- **Gestational Age:** 32 weeks, 4 days
- **Vitals:** BP 132/89 mmHg (Recheck: 129/88) | HR 82 bpm | Temp 36.6°C | Weight 83.0 kg
- **Sonographic Data:** No new scan performed since last visit.

Category	Test	Result	Interpretation
Urinalysis	Protein	Trace	High
	Ketones	Trace	High
	Leukocytes	Negative	Normal
	Nitrites	Negative	Normal
	Glucose	Negative	Normal
Haematology	Haemoglobin	11.97 g/dL	Normal
	Haematocrit	35.2%	Low
	WBC Count	6.57 x10 ⁹ /L	Normal
	RBC Count	3.81 x10 ¹² /L	Normal
	Platelets	223.0 x10 ⁹ /L	Normal
	ANC	5.03 x10 ⁹ /L	Normal
	ALC	1.14 x10 ⁹ /L	Low
Biochemistry	Creatinine	36.74 µmol/L	Low
	ALT	7.23 U/L	Normal
	AST	11.55 U/L	Normal
	ALP	91.32 U/L	Normal
	Total Bilirubin	5.60 µmol/L	Normal
	LDH	162.74 U/L	Low

Visit 4: Follow-up (11/Jun/2026)

The patient presents for her late third-trimester follow-up leading into her scheduled delivery window. Her health status remains compliant and stable. The previous trace proteinuria has completely resolved.

- **Gestational Age:** ~34 weeks, 5 days
- **Vitals:** BP 129/78 mmHg (Recheck: 125/75) | HR 90 bpm | Temp 36.4°C | Weight 82.0 kg
- **Sonographic Data:** No new scan performed since last visit.

Category	Test	Result	Interpretation
Urinalysis	All Parameters	Negative	Normal
Haematology	Haemoglobin	11.60 g/dL	Normal
	Haematocrit	34.0%	Low
	WBC Count	6.50 x10 ⁹ /L	Normal
	RBC Count	3.67 x10 ¹² /L	Low
	Platelets	249.0 x10 ⁹ /L	Normal
	ANC	4.50 x10 ⁹ /L	Normal
	ALC	1.57 x10 ⁹ /L	Normal
Biochemistry	Creatinine	38.01 µmol/L	Low
	ALT	4.95 U/L	Low

Category	Test	Result	Interpretation
	AST	10.0 U/L	Normal
	ALP	109.7 U/L	Normal
	Total Bilirubin	6.38 µmol/L	Normal
	LDH	152.97 U/L	Low

Visit 5: Delivery Visit (29/Jun/2026)

The patient was admitted for an elective Cesarean section indicated by her history of two previous operative deliveries. She delivered a live, healthy infant without any intraoperative or postpartum complications, and recovered normally before being discharged.

- **Gestational Age at Delivery:** 37 weeks, 3 days
- **Vitals:** (Not captured in CTMS for the delivery date).
- **Delivery Outcome:** Elective Cesarean Section (Indication: 2 Previous CS).
- **Maternal Postpartum Status:** Normal recovery. Estimated Blood Loss: 200 mL. Hospital Stay: 2 days (No post-delivery infection or readmission).
- **Fetal/Newborn Assessment Notes:**
 - **Status:** Live, normal birth (Male).
 - **Anthropometry:** Birth Weight 2550g | Length 45cm | Head Circumference 36cm.
 - **Newborn Vitals:** Heart rate 145 bpm | Respiration 41 bpm.
 - **Apgar Scores:** 8 (at 1 min), 9 (at 5 mins).
 - **Clinical Exams:** Physical and neurological examinations were completely normal (eyes, heart, hips, testes, posture, muscle tone, reflexes). No congenital anomalies, birth complications, or neonatal ICU admission.
 - **Feeding:** Breastfeeding established.

(As per the trial protocol, routine blood and urine tests were not mandated or collected for the delivery module.)

Visit 6: Postnatal Follow-up (28/Jul/2026)

Approximately four weeks postpartum, the patient returns for her final study follow-up. Her postnatal clinical recovery has been unremarkable. Blood pressure remains slightly elevated but is completely consistent with her chronic baseline hypertension.

- **Gestational Age:** Postpartum (~4 weeks)
- **Vitals:** BP 138/88 mmHg (Recheck: 136/89) | HR 88 bpm | Temp 36.2°C | Weight 77.0 kg
- **Maternal Status:** Unchanged, clinically stable.
- **Infant Status:** Normal newborn disorder status assessment. No signs of respiratory distress, necrotizing enterocolitis, or intraventricular hemorrhage. No hospitalizations since discharge.

Category	Test	Result	Interpretation
Urinalysis	-	Not Collected	-
Haematology	Haemoglobin	13.8 g/dL	Normal
	Haematocrit	40.8%	Normal
	WBC Count	4.38 x10 ⁹ /L	Normal
	RBC Count	1.3 x10 ¹² /L	Low
	Platelets	271.2 x10 ⁹ /L	Normal
	ANC	2.77 x10 ⁹ /L	Normal
	ALC	1.30 x10 ⁹ /L	Low
Biochemistry	Creatinine	53.2 µmol/L	Normal
	ALT	16.9 U/L	Normal

Category	Test	Result	Interpretation
	AST	14.1 U/L	Normal
	ALP	183.2 U/L	High
	Total Bilirubin	6.93 µmol/L	Normal
	LDH	191.9 U/L	Normal

Final Adjudicator Summary (ISSHP Criteria Focus)

Clinical Course: A 40-year-old multimorbid patient (G4P2) with established chronic hypertension and pre-gestational diabetes successfully completed her pregnancy, delivering a healthy 2550g male infant via planned C-section at 37 weeks. Her chronic conditions were medically managed throughout the trial using Methyldopa, Metformin, dual-therapy Insulin, and Aspirin prophylaxis.

Preeclampsia (PE) Diagnostic Evaluation: Under ISSHP guidelines, adjudicating *superimposed preeclampsia* in a patient with chronic hypertension requires evidence of new-onset maternal organ dysfunction or a sudden, severe exacerbation of baseline hypertension.

1. **Hemodynamics:** While her BP trended upward marginally in the third trimester and postnatally (peaking at 138/89 mmHg), it remained well-controlled on Methyldopa and never breached the severe threshold (≥160/110 mmHg) or the standard diagnostic threshold (≥140/90 mmHg) during clinic vitals checks.

2. **Organ Dysfunction:** The patient maintained excellent baseline renal function (creatinine consistently low/normal) and showed no signs of hepatic injury (ALT/AST normal throughout). Platelets remained robustly normal, ruling out HELLP syndrome.

3. **Proteinuria:** She exhibited only a solitary "Trace" protein result at Visit 3, which resolved entirely by Visit 4.

Adjudicator Verdict: The patient exhibits well-controlled chronic hypertension and diabetes without clinical or laboratory evidence of superimposed preeclampsia or uteroplacental failure during the monitored windows.

User prompt: just give me all the data rows we are taking for each visit as they are before we arrive to the summary

Response: Here are the exact raw data rows pulled from the dataset for MRN 7167, separated by visit, before any summarization or interpretation is applied.

Visit 1: Baseline / Screening (12/May/2026)

Patient Baseline & Medical History

Page Title	Field Label	Data Input	Data Value
Demographics	Age	40	
Demographics	Date of Birth ...	9/16/1985	
Obstetric history	If yes, how many previous pregnancies?	3	
Obstetric history	Number of live births	2	
Obstetric history	Number of miscarriages	1	
Obstetric history	Number of Cesarean sections	2	
Obstetric history	Other reason	Macrosomic baby, hypertension	
Medical History	Comments If applicable	Participant being followed up for Hypertension	
Medical History	Comments If applicable	Methyldopa 500mg po tds, Ranophage 1g po bd, Actrapid 8IU sc tds, Protaphane 12IU sc bd, Asprin 150mg po od	
Family History	Has anyone in participant family ever had hypertension?	Yes	
Family History	Has anyone in participant family ever been diagnosed with diabetes?	Yes	
Pregnancy Assessment	Gestational Age on first USS	15 weeks, 3 days	
Pregnancy Assessment	Current Gestational age (weeks)	30	

Page Title	Field Label	Data Input	Data Value
Pregnancy Assessment	Current Gestational age (days)	3	
Sonographic Data	Fetal Weight	#1 - s:3:"120";	
Sonographic Data	Fetal heart rate	#1 - s:3:"159";	

Vitals & Labs (12/May/2026)

Page Title	Field Label	Data Input
Vital signs	Systolic blood pressure	114
Vital signs	Diastolic blood pressure	78
Vital signs	Body temperature	36.6
Vital signs	Heart Rate	90
Vital signs	Weight	82.8
Urinalysis	Urine leukocytes results	1+
Urinalysis	Urine protein results	Negative
Hematology	Hemoglobin	11.41
Hematology	Hematocrit	33.60
Hematology	White blood cell count	6.40
Hematology	Red blood cell count	3.62
Hematology	Platelets count	251.5
Biochemistry	Creatinine Result	33.7
Biochemistry	Alanine Aminotransferase (ALT) Result	9.2
Biochemistry	Aspartate aminotransferase (AST) Result	11.4
Biochemistry	Alkaline phosphatase (ALP) Result	90.9
Biochemistry	Lactate dehydrogenase (LDH) Result	139.6

Visit 2: Follow-up (21/May/2026)**Vitals, Sonography & Labs (21/May/2026)**

Page Title	Field Label	Data Input
Pregnancy Assessment	Current Gestational Age	31 weeks, 5 days
Sonographic Data	Fetal heart rate	#1 - s:3:"140";
Sonographic Data	Comments	#1 - s:36:"FETAL GROWTH IS WITHIN NORMAL LIMITS";
Vital signs	Systolic blood pressure	128 (Recheck: 127)
Vital signs	Diastolic blood pressure	85 (Recheck: 86)
Vital signs	Heart rate	78
Vital signs	Weight	83
Urinalysis	Urine leukocytes results	Negative
Urinalysis	Urine protein results	Negative
Hematology	Hemoglobin	11.86
Hematology	Hematocrit	34.50

Page Title	Field Label	Data Input
Hematology	White blood cell count	6.58
Hematology	Red blood cell count	3.76
Hematology	Platelets count	240.8
Biochemistry	Creatinine Result	44.24
Biochemistry	Alanine Aminotransferase (ALT) Result	5.38
Biochemistry	Aspartate aminotransferase (AST) Result	11.86
Biochemistry	Alkaline phosphatase (ALP) Result	92.79
Biochemistry	Lactate dehydrogenase (LDH) Result	154.57

Visit 3: Follow-up (27/May/2026)**Vitals & Labs (27/May/2026)**

Page Title	Field Label	Data Input
Pregnancy Assessment	Current Gestational Age	32 weeks, 4 days
Vital signs	Systolic blood pressure	132 (Recheck: 129)
Vital signs	Diastolic blood pressure	89 (Recheck: 88)
Vital signs	Heart rate	82
Vital signs	Weight	83
Urinalysis	Urine ketones results	Trace
Urinalysis	Urine protein results	Trace
Urinalysis	Urine leukocytes results	Negative
Hematology	Hemoglobin	11.97
Hematology	Hematocrit	35.20
Hematology	White blood cell count	6.57
Hematology	Red blood cell count	3.81
Hematology	Platelets count	223
Biochemistry	Creatinine Result	36.74
Biochemistry	Alanine Aminotransferase (ALT) Result	7.23
Biochemistry	Aspartate aminotransferase (AST) Result	11.55
Biochemistry	Alkaline phosphatase (ALP) Result	91.32
Biochemistry	Lactate dehydrogenase (LDH) Result	162.74

Visit 4: Follow-up (11/Jun/2026)**Vitals & Labs (11/Jun/2026)**

Page Title	Field Label	Data Input
Pregnancy Assessment	Current Gestational Age	~34 weeks, 5 days
Vital signs	Systolic blood pressure	129 (Recheck: 125)
Vital signs	Diastolic blood pressure	78 (Recheck: 75)
Vital signs	Heart rate	90
Vital signs	Weight	82

Page Title	Field Label	Data Input
Urinalysis	Urine protein results	Negative
Hematology	Hemoglobin	11.60
Hematology	Hematocrit	34.00
Hematology	White blood cell count	6.50
Hematology	Red blood cell count	3.67
Hematology	Platelets count	249
Biochemistry	Creatinine Result	38.01
Biochemistry	Alanine Aminotransferase (ALT) Result	4.95
Biochemistry	Aspartate aminotransferase (AST) Result	10.0
Biochemistry	Alkaline phosphatase (ALP) Result	109.70
Biochemistry	Lactate dehydrogenase (LDH) Result	152.97

Visit 5: Delivery Visit (29/Jun/2026)

Delivery & Newborn Assessment (29/Jun/2026)

Page Title	Field Label	Data Input
Delivery Outcome	Type of delivery	Cesarean
Delivery Outcome	Reason for Cesarean section	2 Previous CS
Delivery Outcome	Gestational age at delivery	37 weeks, 3 days
Delivery Outcome	Estimated blood loss	200
Delivery Outcome	Did participant experience infection post-delivery?	No
Delivery Outcome	How many days did participant stay in hospital	2
Newborn Assessment	Birth Weight	2550
Newborn Assessment	Length	45
Newborn Assessment	Head Circumference	36
Newborn Assessment	Heart rate	145
Newborn Assessment	Respiration	41
Newborn Assessment	Apgar Score at 1 Minute	8
Newborn Assessment	Apgar Score at 5 Minutes	9
Newborn Assessment	Physical Examination	Normal (Eyes, Heart, Hips)
Newborn Assessment	Neurological Examination	Normal (Posture, Tone, Reflexes)
Newborn Assessment	Congenital Anomalies	No

Visit 6: Postnatal Follow-up (28/Jul/2026)

Vitals & Labs (28/Jul/2026)

Page Title	Field Label	Data Input
Newborn Assessment	Date of newborn disorder status assessment	7/28/2026 8:49:00
Newborn Assessment	Hospitalized?	No
Vital signs	Systolic blood pressure	138 (Recheck: 136)
Vital signs	Diastolic blood pressure	88 (Recheck: 89)

Page Title	Field Label	Data Input
Vital signs	Pulse	88
Vital signs	Weight	77
Hematology	Hemoglobin	13.8
Hematology	Hematocrit	40.8
Hematology	White blood cell count	4.38
Hematology	Red blood cell count	1.3
Hematology	Platelets count	271.2
Biochemistry	Creatinine Result	53.2
Biochemistry	Alanine Aminotransferase (ALT) Result	16.9
Biochemistry	Aspartate aminotransferase (AST) Result	14.1
Biochemistry	Alkaline phosphatase (ALP) Result	183.2
Biochemistry	Lactate dehydrogenase (LDH) Result	191.9

User prompt: not a summary its for my code editor to know the changes i am making to the data mappng and presentation

```
Response: {
  "data_mapping_schema": {
    "demographics": {
      "Age": "age",
      "Date of Birth ...": "dob",
      "If yes, how many previous pregnancies?": "gravida",
      "Number of live births": "para_live",
      "Number of miscarriages": "para_miscarriage",
      "Number of Cesarean sections": "previous_cs"
    },
    "vitals": {
      "Systolic blood pressure": "bp_systolic",
      "Diastolic blood pressure": "bp_diastolic",
      "Systolic blood pressure recheck": "bp_systolic_recheck",
      "Diastolic blood pressure recheck": "bp_diastolic_recheck",
      "Heart Rate": "heart_rate",
      "Pulse": "heart_rate",
      "Body temperature": "temperature",
      "Weight": "weight"
    },
    "urinalysis": {
      "Urine blood results": "ua_blood",
      "Urine ketones results": "ua_ketones",
      "Urine leukocytes results": "ua_leukocytes",
      "Urine nitrites results": "ua_nitrites",
      "Urine protein results": "ua_protein",
      "Urine glucose results": "ua_glucose"
    },
    "haematology": {
      "Hemoglobin": "haemoglobin",
      "Hematocrit": "haematocrit",
      "White blood cell count": "wbc",
      "Red blood cell count": "rbc",
      "Platelets count": "platelets",
      "Absolute neutrophil count": "anc",
      "Absolute lymphocytes count": "alc"
    },
    "biochemistry": {
      "Creatinine Result": "creatinine",
      "Alanine Aminotransferase (ALT) Result": "alt",
      "Aspartate aminotransferase (AST) Result": "ast",
      "Alkaline phosphatase (ALP) Result": "alp",
      "Total Bilirubin Result": "total_bilirubin",
      "Lactate dehydrogenase (LDH ) Result": "ldh"
    }
  }
}
```

```

    },
    "obstetrics_and_newborn": {
      "Gestational Age on first USS": "ega_baseline",
      "Current Gestational age (weeks)": "ega_weeks",
      "Current Gestational age (days)": "ega_days",
      "Gestational age at delivery": "ega_delivery",
      "Type of delivery": "delivery_mode",
      "Estimated blood loss": "ebl_ml",
      "Birth Weight": "newborn_weight_g",
      "Apgar Score at 1 Minute": "apgar_1m",
      "Apgar Score at 5 Minutes": "apgar_5m"
    }
  },
  "raw_visit_payloads": {
    "MRN_7167": [
      {
        "visit_id": "V1",
        "visit_date": "2026-05-12",
        "vitals": {
          "bp_systolic": 114,
          "bp_diastolic": 78,
          "heart_rate": 90,
          "temperature": 36.6,
          "weight": 82.8
        },
        "labs": {
          "ua_leukocytes": "1+",
          "ua_protein": "Negative",
          "haemoglobin": 11.41,
          "haematocrit": 33.60,
          "wbc": 6.40,
          "rbc": 3.62,
          "platelets": 251.5,
          "creatinine": 33.7,
          "alt": 9.2,
          "ast": 11.4,
          "alp": 90.9,
          "ldh": 139.6
        }
      },
      {
        "visit_id": "V2",
        "visit_date": "2026-05-21",
        "vitals": {
          "bp_systolic": 128,
          "bp_diastolic": 85,
          "bp_systolic_recheck": 127,
          "bp_diastolic_recheck": 86,
          "heart_rate": 78,
          "temperature": 36.7,
          "weight": 83.0
        },
        "labs": {
          "ua_leukocytes": "Negative",
          "ua_protein": "Negative",
          "haemoglobin": 11.86,
          "haematocrit": 34.50,
          "wbc": 6.58,
          "rbc": 3.76,
          "platelets": 240.8,
          "creatinine": 44.24,
          "alt": 5.38,
          "ast": 11.86,
          "alp": 92.79,
          "ldh": 154.57
        }
      },
      {
        "visit_id": "V3",
        "visit_date": "2026-05-27",
        "vitals": {
          "bp_systolic": 132,
          "bp_diastolic": 89,
          "bp_systolic_recheck": 129,

```

```

      "bp_diastolic_recheck": 88,
      "heart_rate": 82,
      "temperature": 36.6,
      "weight": 83.0
    },
    "labs": {
      "ua_ketones": "Trace",
      "ua_protein": "Trace",
      "ua_leukocytes": "Negative",
      "haemoglobin": 11.97,
      "haematocrit": 35.20,
      "wbc": 6.57,
      "rbc": 3.81,
      "platelets": 223.0,
      "creatinine": 36.74,
      "alt": 7.23,
      "ast": 11.55,
      "alp": 91.32,
      "ldh": 162.74
    }
  },
  {
    "visit_id": "V4",
    "visit_date": "2026-06-11",
    "vitals": {
      "bp_systolic": 129,
      "bp_diastolic": 78,
      "bp_systolic_recheck": 125,
      "bp_diastolic_recheck": 75,
      "heart_rate": 90,
      "temperature": 36.4,
      "weight": 82.0
    },
    "labs": {
      "ua_protein": "Negative",
      "haemoglobin": 11.60,
      "haematocrit": 34.00,
      "wbc": 6.50,
      "rbc": 3.67,
      "platelets": 249.0,
      "creatinine": 38.01,
      "alt": 4.95,
      "ast": 10.0,
      "alp": 109.70,
      "ldh": 152.97
    }
  },
  {
    "visit_id": "V5",
    "visit_date": "2026-06-29",
    "delivery_metrics": {
      "delivery_mode": "Cesarean",
      "indication": "2 Previous CS",
      "ega_delivery": "37 weeks, 3 days",
      "ebl_ml": 200,
      "hospital_days": 2,
      "newborn_weight_g": 2550,
      "apgar_1m": 8,
      "apgar_5m": 9
    }
  },
  {
    "visit_id": "V6",
    "visit_date": "2026-07-28",
    "vitals": {
      "bp_systolic": 138,
      "bp_diastolic": 88,
      "bp_systolic_recheck": 136,
      "bp_diastolic_recheck": 89,
      "heart_rate": 88,
      "weight": 77.0
    },
    "labs": {
      "haemoglobin": 13.8,

```

```

        "haematocrit": 40.8,
        "wbc": 4.38,
        "rbc": 1.3,
        "platelets": 271.2,
        "creatinine": 53.2,
        "alt": 16.9,
        "ast": 14.1,
        "alp": 183.2,
        "ldh": 191.9
    }
}
]
}
}

```

User prompt: is this complete for all visits it seems small, i have uploaded the previous mapping file for editing (this is very important that you do not make mistakes if you need clarity ask before messing up a global project)

Response: I have extracted all the exact raw data rows for MRN 7167 that were used to build the summaries for Visits 1 through 6.

Since the raw dataset is 1,304 rows long, I have dumped the complete, unedited subset into a text file so you can easily review the precise mappings in your code editor.

code-generated-file-0340b0b7-d3f2-4640-ad6e-e7f307998adb [generated with file_gen tool]

This file contains exactly what the site coordinators entered into the CTMS (the Page Title, Field Label, Data Input, and Data Value), in its raw state prior to any mapping or interpretation.

User prompt: write an updated py file

Response: Here is the updated realtime_mapping.py script. It integrates the exact data mapping schema we developed, utilizing the exact Field Label strings found in your CTMS raw data export to accurately parse vitals, labs, delivery metrics, and newborn assessments into clean, standardized variables.

```

"""Controlled RealTime composite mapping and privacy classification."""
import re
from datetime import datetime, timezone

MAPPING_VERSION = "RT-MAP-2.0"

# Updated DIRECT_ALIASES to support the new standardized schema mapping
DIRECT_ALIASES = {
    "event_dt": "visit_date", "visit_date": "visit_date",
    "ga_event": "ega_weeks", "sbp": "bp_systolic", "dbp": "bp_diastolic",
    "plt": "platelets", "upcr": "upcr"
}

PROHIBITED_PATTERNS = tuple(re.compile(p,re.I) for p in (
    r"s\s*flt[- ]?1", r"plgf", r"seng", r"biomarker", r"poc result",
    r"point.of.care", r"treatment allocation", r"randomi[sz]ation allocation", r"circa.?red",
))

DIRECT_IDENTIFIER_PATTERNS = tuple(re.compile(p,re.I) for p in (
    r"\bmrn\b", r"screening #", r"randomization #", r"randomisation #",
    r"date of birth", r"\bptid\b"
))

RESTRICTED_PATTERNS = tuple(re.compile(p,re.I) for p in (
    r"electronic signature", r"research nurse", r"audit trail", r"file upload", r"reviewer"
))

# Updated RULES with new CTMS mapping targets
RULES = [
    # Demographics & Obstetric History
    ("age", (r"^\d+$",)),
    ("dob", (r"date of birth",)),
    ("gravida", (r"how many previous\s*pregnancies",)),
    ("para_live", (r"number of live births",)),
    ("para_miscarriage", (r"number of miscarriages",)),
    ("previous_cs", (r"number of cesarean sections",)),

```

```

# Visit & Dating
("visit_date", (r"visit date", r"date of visit", r"date of the visit")),
("ega_baseline", (r"gestational age on first uss",)),
("ega_weeks", (r"current gestational age.*weeks", r"\bga_weeks\b")),
("ega_days", (r"current gestational age.*days", r"\bga_days\b")),

# Vitals
("bp_systolic", (r"^systolic blood pressure$", r"\bsystolic\b")),
("bp_diastolic", (r"^diastolic blood pressure$", r"\bdiastolic\b")),
("bp_systolic_recheck", (r"systolic blood pressure recheck",)),
("bp_diastolic_recheck", (r"diastolic blood pressure recheck",)),
("heart_rate", (r"^heart rate$", r"\bpulse\b")),
("temperature", (r"body temperature",)),
("weight", (r"^weight$",)),

# Urinalysis
("ua_blood", (r"urine blood results",)),
("ua_ketones", (r"urine ketones results",)),
("ua_leukocytes", (r"urine leukocytes results",)),
("ua_nitrites", (r"urine nitrites results",)),
("ua_protein", (r"urine protein results", r"dipstick.*result")),
("ua_glucose", (r"urine glucose results",)),
("upcr", (r"\bupcr\b.*result", r"protein.?creatinine ratio result")),

# Haematology
("haemoglobin", (r"^hemoglobin$", r"^haemoglobin$")),
("haematocrit", (r"^hematocrit$", r"^haematocrit$")),
("wbc", (r"white blood cell count",)),
("rbc", (r"red blood cell count",)),
("platelets", (r"platelets count", r"\bplatelets\b")),
("anc", (r"absolute neutrophil count",)),
("alc", (r"absolute lymphocytes count",)),

# Biochemistry
("creatinine", (r"creatinine result", r"\bcreatinine\b")),
("alt", (r"alanine aminotransferase", r"\balt\b")),
("ast", (r"aspartate aminotransferase", r"\bast\b")),
("alp", (r"alkaline phosphatase", r"\balp\b")),
("total_bilirubin", (r"total bilirubin result", r"\bbilirubin\b")),
("ldh", (r"lactate dehydrogenase", r"\bldh\b")),

# Delivery Outcome
("delivery_mode", (r"type of delivery",)),
("indication", (r"reason for cesarean section",)),
("ega_delivery", (r"gestational age at delivery",)),
("ebl_ml", (r"estimated blood loss",)),
("hospital_days", (r"how many days did participant stay in hospital",)),

# Newborn Assessment
("newborn_weight_g", (r"birth weight",)),
("apgar_1m", (r"apgar score at 1 minute",)),
("apgar_5m", (r"apgar score at 5 minutes",)),

# Additional ConMeds/AE Logic
("medication_name", (r"medication name", r"drug name", r"concomitant medication")),
("medication_dose", (r"medication dose", r"\bdose\b")),
]

RULES = [(canonical, tuple(re.compile(p,re.I) for p in patterns)) for canonical, patterns in RULES]

def norm(value):
    return re.sub(r"\s+", " ", str(value or "").strip().lower())

def metadata_text(row):
    return " | ".join(norm(row.get(k)) for k in ("Form Title", "Form Version", "Page Title", "Field Label", "Field type"))

def classify(row):
    text = metadata_text(row)
    if any(p.search(text) for p in PROHIBITED_PATTERNS): return "PROHIBITED_BLIENDED"
    if any(p.search(text) for p in DIRECT_IDENTIFIER_PATTERNS): return "DIRECT_IDENTIFIER"
    if any(p.search(text) for p in RESTRICTED_PATTERNS): return "RESTRICTED_OPERATIONAL_METADATA"
    return "PERMITTED_CLINICAL_EVIDENCE" if map_variable(row) else "UNMAPPED"

def map_variable(row):

```

```

export_name = norm(row.get("Export Variable Name")).replace(" ", "_")
if export_name in DIRECT_ALIASES: return DIRECT_ALIASES[export_name]
text = " | ".join(norm(row.get(k)) for k in ("Page Title", "Field Label", "Export Variable Name"))
for canonical, patterns in RULES:
    if any(p.search(text) for p in patterns): return canonical
return None

def source_value(row):
    return (row.get("Data Value") or row.get("Data Input") or "").strip()

def parse_datetime(value):
    v = (value or "").strip().split("|")[0].strip()
    if not v: return None
    normalized = re.sub(r"\s+(?:SAST|SAT|UTC|GMT|CAT)$", "", v, flags=re.I).strip()
    try:
        parsed = datetime.fromisoformat(normalized.replace("Z", "+00:00"))
        return parsed.astimezone(timezone.utc).replace(tzinfo=None) if parsed.tzinfo else parsed
    except ValueError:
        pass
    for fmt in (
        "%m/%d/%Y %H:%M:%S", "%m/%d/%Y %H:%M", "%m/%d/%Y",
        "%d/%m/%Y %H:%M:%S", "%d/%m/%Y %H:%M", "%d/%m/%Y",
        "%d/%b/%Y %H:%M:%S", "%d/%b/%Y %H:%M",
        "%d/%b/%Y %I:%M:%S %p", "%d/%b/%Y %I:%M %p", "%d/%b/%Y",
        "%Y-%m-%d %H:%M:%S", "%Y-%m-%d %H:%M", "%Y-%m-%d",
        "%d-%b-%Y %H:%M:%S", "%d-%b-%Y %H:%M", "%d-%b-%Y",
    ):
        try: return datetime.strptime(normalized, fmt)
        except ValueError: pass
    return None

def parse_numeric(value):
    m = re.search(r"[-+]?[d+](?:\.\d+)?", str(value or "").replace(",", ""))
    try: return float(m.group()) if m else None
    except ValueError: return None

def parse_coded(value):
    v = norm(value)
    if v in {"yes", "y", "true", "1"}: return "YES"
    if v in {"no", "n", "false", "0"}: return "NO"
    if "not done" in v: return "NOT_DONE"
    if "not applicable" in v: return "NOT_APPLICABLE"
    if "unknown" in v: return "UNKNOWN"
    if "not answered" in v: return "MISSING"
    return str(value or "").strip() or None

def visit_code(form_title):
    text = norm(form_title)
    if "screening" in text or "v01" in text: return "V01", 1, "SCHEDULED"
    m = re.search(r"visit\s*(\d+)", text)
    if m: return f"V{int(m.group(1)):02d}", int(m.group(1)), "SCHEDULED"
    if "unscheduled" in text: return "UNSCHEDULED", 90, "UNSCHEDULED"
    if "early termination" in text: return "EARLY_TERMINATION", 95, "EARLY_TERMINATION"
    if "adverse event" in text: return "ADVERSE_EVENT", 96, "EVENT"
    if "protocol deviation" in text: return "PROTOCOL_DEVIATION", 97, "EVENT"
    return "OTHER", 99, "OTHER"

```

User prompt: summary of the changes to mapping

Response: Here is the summary of the changes made to `realtime_mapping.py` to upgrade it to version RT-MAP-2.0 based on your CTMS data mapping schema:

1. Version Update

- Bumped `MAPPING_VERSION` from "RT-MAP-1.0" to "RT-MAP-2.0".

2. Direct Aliases Expansion (DIRECT_ALIASES)

- Added mappings for core clinical indicators to ensure direct variable resolution (sbp -> bp_systolic, dbp -> bp_diastolic, plt -> platelets).

3. Comprehensive Rule Set Modernization (RULES)

- **Demographics & Obstetric History:** Added explicit canonical rules for patient age (`age`), date of birth (`dob`), gravida (`gravida`), live births (`para_live`), miscarriages (`para_miscarriage`), and previous Cesarean sections (`previous_cs`).
 - **Vitals Tracking:** Expanded regex matching to accurately capture exact Field Labels for systolic/diastolic blood pressure (including rechecks), heart rate, temperature, and weight.
 - **Urinalysis & Chemistry:** Standardized urine parameters (`ua_blood`, `ua_ketones`, `ua_leukocytes`, `ua_nitrites`, `ua_protein`, `ua_glucose`) alongside comprehensive biochemistry panels (`creatinine`, `alt`, `ast`, `alp`, `total_bilirubin`, `ldh`).
 - **Haematology Indices:** Integrated precise matching for haemoglobin, haematocrit, WBC, RBC, platelets, absolute neutrophil count (ANC), and absolute lymphocyte count (ALC).
 - **Delivery & Newborn Metrics:** Added mappings for delivery outcomes (`delivery_mode`, `indication`, `ega_delivery`, `eb1_ml`, `hospital_days`) and newborn health assessments (`newborn_weight_g`, `apgar_1m`, `apgar_5m`).
-